

ID No: _____ Name: _____ Sec No.: _____

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI

II SEMESTER 2025-26

CS/ECE/EEE/INSTR F241 Microprocessor Programming & Interfacing

22-04-2026 (Wednesday)

QUIZ-6 (closed book)

A

Max time: 10 Min

Max Marks: 08

Q1. An 8086 system uses 8259A in cascaded mode with: 1 Master PIC , 2 Slave PICs . The connections are: Each 8259 can handle 8 interrupts (IR0 to IR7)

- Slave 1 is connected to IR2 of the master, Slave 2 is connected to IR5 of the master

The vector base addresses are initialized as follows:

- Master = 40H , Slave 1 = 70H, Slave 2 = A0H

An interrupt occurs on IR3 of Slave 2

Answer the following:

(a)What is the interrupt vector number for IR3 of Slave 2

(b)What is the starting physical address location in the interrupt vector table from which the ISR address will be fetched for IR1 of the master

(c) What will be the CAS0–CAS2 signals placed by the Master during the interrupt acknowledge cycle?

Question	Answer
1. a[2M]	A3H
1.b[2M]	104H
1.c [2M]	101

Q.2. An Intel 8254 Programmable Interval Timer is programmed in Mode 3. The counter C0 is loaded with a value of 5000 (decimal). Input to CLK0 is 2.5 Mhz. What is the duration of the high period (ms) of the output? The gate signal is maintained high throughout. [2M]

Ans: 1ms

Show Calculation:

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B

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Q1. An 8086 system uses 8259A in cascaded mode with: 1 Master PIC , 2 Slave PICs . The connections are: Each 8259 can handle 8 interrupts (IR0 to IR7)

- Slave 1 is connected to IR2 of the master, Slave 2 is connected to IR5 of the master

The vector base addresses are initialized as follows:

- Master = 40H , Slave 1 = 70H, Slave 2 = A0H

An interrupt occurs on IR3 of Slave 1

Answer the following:

(a)What is the interrupt vector number for IR3 of Slave 1

(b)What is the starting physical address location in the interrupt vector table from which the ISR address will be fetched for IR2 of the master

(c) What will be the CAS0–CAS2 signals placed by the Master during the interrupt acknowledge cycle?

Question	Answer
1. a [2M]	73h
1.b[2M]	108
1.c [2M]	010

Q.2. An Intel 8254 Programmable Interval Timer is programmed in Mode 3. The counter C0 is loaded with a value of 5000 (decimal). Input to CLK0 is 2.0 Mhz. What is the duration of the high period (ms) of the output? The gate signal is maintained high throughout. [2M]

Ans:

Show Calculation:

